TurboPrefill: Alternative Multi-GPU Prefill Scheduling

Track: Physical AI

Project background

TurboPrefill is an experimental scheduling method for multi-GPU LLM inference in llama.cpp. It targets long-context prefill workloads where GPU utilization may become uneven because host-device transfers and computation are not efficiently overlapped.

Architecture

TurboPrefill changes only the scheduling strategy. Complete ubatches are accumulated in host memory and dispatched through an alternative execution pipeline that coordinates H2D/D2H transfers and GPU computation. The numerical computation itself is unchanged.

Validation

A dedicated validation fork outputs the hash of the complete first-token logits array together with the Top-16 logits after every prefill. Running identical models with TurboPrefill=0 and TurboPrefill=1 allows numerical verification that the first-token logits remain identical.

Current status

- CUDA implementation completed.
- Validation framework completed.
- Tested on multiple NVIDIA multi-GPU configurations.
- ROCm adaptation planned.
- Submitted as an architectural proof of concept for AMD GPU systems.

Repositories

Main project:
https://github.com/sergey-automation/TurboPrefill

Validation fork:
https://github.com/sergey-automation/llama.cpp/tree/b1dd9d99cf9609f90569881989e524219ac32c98

Related work

TurboPrefill also participated in OpenAI Build Week.

TurboPrefill by Trykhlieb adapted for ik_llama.cpp by Ivan Kawrakow.

https://devpost.com/software/turboprefill